All Dice, No Slice: Metric Artifacts, Data Leakage, and Task Interference in Multi-Task Computational Pathology

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Abstract. Simultaneous classification and lesion segmentation via hard parameter-sharing U-Nets is an appealing paradigm for computational pathology. Recently, classification accuracies of $82\% - 90\%$ and Dice scores of $97\% - 99\%$ were reported across diverse medical imaging modalities using standard convolutional backbones and static loss weighting. We conduct a 26-experiment replication and methodological audit across four datasets: TCGA-LGG brain-tumor MRI, PANDA prostate biopsies, SIIM-ACR pneumothorax radiographs, and the 19-tissue PanNuke benchmark. Under strict patient-level data isolation (`GroupKFold`), the claimed accuracies are not reproducible, and the reported Dice levels are attainable only as empty-mask artifacts: all four of our SIIM configurations converge to all-empty predictions yet report 77.74% Dice — exactly the lesion-free fraction of the validation split under the convention that scores empty-over-empty slices as perfect agreement. On fine-grained 6-class prostate cancer grading (PANDA), empirical top-1 accuracy spans $29.04\% - 46.15\%$ and Dice spans $28.94\% - 44.34\%$ across all 12 evaluated configurations, with static 5:1-weighted 10^{-3} baselines (Runs 03–04) reaching $45.15\% - 46.15\%$ / $43.30\% - 44.34\%$ (vs. 88.0% and 99.0% claimed). We trace the discrepancy to three systemic causes: (1) empty-mask Dice crediting, (2) patient-level data leakage, and (3) task interference from the joint optimization recipe. Matched ablations further show that skip connections are a null result, whereas removing Macenko stain normalization *improves* fine-grained classification (+5.51% on PANDA, +2.68% on PanNuke). We release our benchmark and audit protocol to prevent such metric artifacts in future multi-task pathology studies.

Keywords: Computational Pathology · Multi-Task Learning · U-Net · Reproducibility · Metric Artifacts · Data Leakage · GradNorm

1 Introduction

Deep learning systems in digital pathology and radiology are increasingly required to deliver both diagnostic categorization and lesion delineation [10,6].

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Large visual foundation models (e.g., UNI [7], CONCH [11], and Virchow [19]) show that clinically actionable representations require massive pretraining and self-supervised objectives across gigapixel whole-slide images (WSIs). In resource-constrained deployments, hard parameter sharing via an encoder–decoder backbone (a U-Net [15] with dual prediction heads) remains a lightweight alternative that couples both objectives in a single forward pass.

Recently, Rhanoui et al. [14] claimed that a standard U-Net with lightweight encoders (VGG16 [17] and MobileNetV2 [16]) trained with static loss weights ($\lambda_{seg} = 5.0, \lambda_{cls} = 1.0$) and a high learning rate ($\eta = 10^{-3}$) simultaneously solves both classification and segmentation across four distinct medical modalities (Brain MRI, Skin Dermoscopy, Prostate Histopathology, and Chest X-Ray), reporting $82\% - 90\%$ classification accuracy and $97\% - 99\%$ Dice similarity coefficients.

In this paper, we report a 26-run replication campaign and systematic methodological audit. Our findings demonstrate that:

1. **Empirical Irreproducibility:** When evaluated with strict patient/biopsy-level isolation (`GroupKFold`), multi-class grading on PANDA prostate biopsies collapses from the claimed 88.0% accuracy / 99.0% Dice down to an empirical range of $29.04\% - 46.15\%$ Acc / $28.94\% - 44.34\%$ Dice across all 12 configurations, with the static 5:1-weighted 10^{-3} baselines (Runs 03–04) achieving $45.15\% - 46.15\%$ / $43.30\% - 44.34\%$.
2. **The Empty-Mask Dice Metric Artifact:** Near-perfect (99%) Dice claims on sparse-lesion datasets such as SIIM-ACR pneumothorax can arise from crediting lesion-free slices ($|Y| = |\hat{Y}| = 0$) as Dice = 1.0; our four SIIM runs measure exactly this floor (77.74%) under an all-empty predictor.
3. **Controlled Multi-Task & Ablation Analysis:** Through strictly matched factorial controls (learning-rate pairs, static loss-weight ratios $\lambda_{seg} : \lambda_{cls} \in \{1{:}10,\ 1{:}1, 5{:}1, 10{:}1\}$, GradNorm [8] dynamic balancing, Macenko normalization, and skip-connection ablations), we isolate how individual components govern multi-task optimization: dynamic gradient balancing actively impairs fine-grained classification ($45.39\% \rightarrow 29.04\%$ Acc), while skip connections are a verified null result.

To facilitate complete reproducibility, all training scripts, raw JSON summaries for all 26 runs, evaluation checkpoints, and generated figures are publicly available at https://github.com/ApatheticMioz/cancer-pathology-dl [1].

2 Multi-Task Formulation & Evaluation Metrics

2.1 Architectural Formulation

Let $\mathbf{x} \in \mathbb{R}^{H \times W \times C}$ denote an input image tile. A shared convolutional encoder $f_\theta(\mathbf{x})$ extracts a bottleneck latent representation $\mathbf{z} \in \mathbb{R}^{h \times w \times d}$. The classification branch applies global average pooling (GAP) followed by a two-layer multilayer perceptron (hidden width 256, dropout 0.5):

$$\hat{\mathbf{y}}_{cls} = \mathrm{Softmax}(\mathbf{W}_2\, \phi(\mathbf{W}_1\, \mathrm{GAP}(\mathbf{z}) + \mathbf{b}_1) + \mathbf{b}_2) \in \Delta^{K-1}, \tag{1}$$

where K is the number of diagnostic classes, Δ^{K-1} denotes the probability simplex, and $\phi(\cdot)$ is the ReLU nonlinearity. The segmentation branch passes $\mathbf{z}$ through a decoder Dec_ϕ with lateral skip connections $\{\mathbf{s}_l\}_{l=1}^{L}$ from encoder stages, yielding unnormalized pixel logits $\mathbf{l}_{seg} \in \mathbb{R}^{H\times W\times M}$. For binary segmentation ($M = 1$), $\hat{\mathbf{y}}_{seg} = \sigma(\mathbf{l}_{seg})$ with $\sigma(\cdot)$ the sigmoid; for $M > 1$, class probabilities are obtained via channel-wise Softmax.

The joint multi-task training objective is defined as:

$$\mathcal{L}_{total}(\theta, \phi, \mathbf{W}_{cls}) = \lambda_{cls}\mathcal{L}_{cls}(\mathbf{y}_{cls}, \hat{\mathbf{y}}_{cls}) + \lambda_{seg}\mathcal{L}_{seg}(\mathbf{y}_{seg}, \mathbf{l}_{seg}), \tag{2}$$

where $\mathcal{L}_{cls}$ is categorical cross-entropy $\mathcal{L}_{CE}$. In the published study of Rhanoui et al. [14], the segmentation loss was stated as $\mathcal{L}_{seg} = \mathcal{L}_{BCE} + \mathcal{L}_{Dice}$; the implementation audited for this study, however, optimizes purely cross-entropy objectives, without any Dice-based penalty term (a documented deviation):

$$\mathcal{L}_{seg}(\mathbf{y}_{seg}, \mathbf{l}_{seg}) = \begin{cases} \mathcal{L}_{BCE}(\mathbf{y}_{seg}, \mathbf{l}_{seg}), & M = 1 \text{ (binary tasks)}, \\ \mathcal{L}_{CE}(\mathbf{y}_{seg}, \mathbf{l}_{seg}), & M > 1 \text{ (multi-class tasks)}. \end{cases} \tag{3}$$

The Dice similarity coefficient (DSC) is evaluated post-hoc as a validation metric, not as an active loss term during gradient descent.

2.2 The Empty-Mask Dice Metric Artifact

The Sørensen–Dice coefficient between ground-truth mask Y and binary prediction $\hat{Y}$ ($Y, \hat{Y} \in \{0,1\}^{H\times W}$) is:

$$\mathrm{Dice}(Y, \hat{Y}) = \frac{2|Y \cap \hat{Y}|}{|Y| + |\hat{Y}| + \epsilon}, \quad \epsilon = 10^{-8}; \;\; \mathrm{Dice} := 1 \text{ if } |Y| + |\hat{Y}| = 0. \tag{4}$$

Empty references and predictions are recognized validation pitfalls in biomedical image analysis [13]; two conventions govern empty slices:

- **Foreground-Only Evaluation:** Compute Dice strictly over non-empty target slices ($\{i \mid |Y_i| > 0\}$).
- **Empty-Credit Assignment:** Assign Dice $= 1.0$ when $|Y| = 0 \wedge |\hat{Y}| = 0$, and Dice $= 0.0$ when $|Y| = 0 \wedge |\hat{Y}| > 0$.

Averaged over all slices under empty-credit assignment, the macroscopic mean is

$$\overline{\mathrm{Dice}}_{\mathrm{all}} = \tfrac{1}{N}\textstyle\sum_{i=1}^{N} d_i, \;\; d_i = \mathrm{Dice}(Y_i, \hat{Y}_i) \text{ if } |Y_i| > 0, \text{ else } \mathbf{1}\Big[|\hat{Y}_i| = 0\Big], \tag{5}$$

which for mean foreground Dice $\overline{\mathrm{Dice}}_{\mathrm{fg}}$ on lesion-bearing slices is the convex combination

$$\overline{\mathrm{Dice}}_{\mathrm{all}} = \rho + (1-\rho)\cdot\overline{\mathrm{Dice}}_{\mathrm{fg}}, \tag{6}$$

where ρ is the proportion of lesion-free slices. In sparse pathologies such as SIIM-ACR pneumothorax, a model with a genuine foreground Dice of 77.74% would, as an analytic illustration, report $\overline{\mathrm{Dice}}_{\mathrm{all}} = 0.777 \times 1.0 + 0.223 \times 0.7774 = 95.04\%$

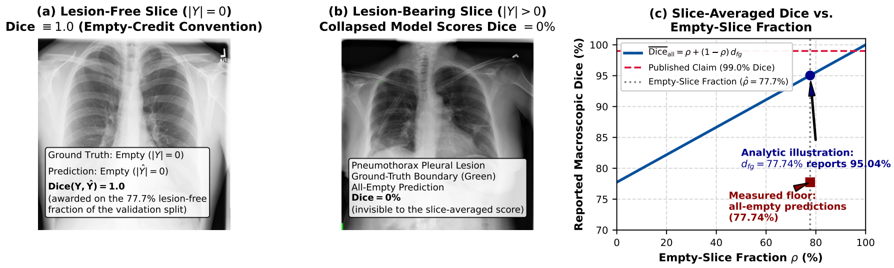

Fig. 1: The empty-mask Dice artifact on SIIM-ACR pneumothorax. (a) Lesion-free slice ($|Y| = 0$): under empty-credit assignment an all-empty prediction is awarded Dice $= 1.0$; lesion-free slices constitute $\hat{\rho} = 77.7\%$ of our validation split. (b) Lesion-bearing slice ($|Y| > 0$, ground-truth boundary in green): our collapsed models predict empty masks here and score Dice $= 0$, invisible to the slice-averaged mean. (c) Convex-combination curve (Eq. 6): a foreground Dice of 77.74% would report 95.04% at $\hat{\rho} = 0.777$ (analytic illustration), whereas all four of our SIIM runs measure the all-empty floor of 77.74% itself.

at the observed lesion-free fraction $\hat{\rho} = 1{,}659/2{,}135 = 0.777$. Our four SIIM runs realize the more extreme failure mode: emitting empty masks on every validation slice throughout training, each reported 77.74% (Runs 05, 06, 11, 12) *is the all-empty floor itself* — Eq. 5 with zero foreground Dice. As Fig. 1 shows, slice-averaged Dice on sparse data reports the lesion-free fraction rather than lesion overlap, so published 99.0% Dice claims [14] are uninterpretable as evidence of lesion localization.

2.3 Adaptive Gradient Balancing via GradNorm

To evaluate whether dynamic loss balancing mitigates cross-task gradient competition, we introduced a parameterized variant of GradNorm [8] in Phase v2.

Let W denote the shared convolutional filters and normalization layers of the U-Net encoder f_θ. The L_2 gradient norm of task $i \in \{seg, cls\}$ with respect to W at minibatch t is:

$$G_W^{(i)}(t) = \left\| \nabla_W w_i(t) \mathcal{L}_i(t) \right\|_2 . \tag{7}$$

For stability in the primary `Adam` optimizer, balancing weights are learnable log-weights $\mu_i(t) = \ln w_i(t)$ (initialized to $w_{seg}(0) = 5.0$, $w_{cls}(0) = 1.0$), updated jointly with the network parameters. Minibatch optimization minimizes the L_1 gradient disparity objective:

$$\mathcal{L}_{grad}(t) = \sum_{i \in \{seg, cls\}} \left| G_W^{(i)}(t) \; - \; \bar{G}_W(t) \cdot [r_i(t)]^{\alpha} \right| , \tag{8}$$

where $\bar{G}_W(t) = \frac{1}{2} \sum_j G_W^{(j)}(t)$ is the average gradient norm across tasks, $\alpha = 1.5$ is the asymmetry parameter, and $r_i(t)$ represents the relative inverse training rate:

$$r_i(t) = \frac{\tilde{\mathcal{L}}_i(t)}{\frac{1}{2}\sum_{j\in\{seg,cls\}}\tilde{\mathcal{L}}_j(t)}, \quad \text{with} \quad \tilde{\mathcal{L}}_i(t) = \frac{\mathcal{L}_i(t)}{\mathcal{L}_i(0)}, \tag{9}$$

where $\mathcal{L}_i(0)$ is the initial step-0 loss. Following Chen et al. [8], slower-training tasks exhibit higher loss ratios $r_i(t) > 1$, raising their target gradient norm $\bar{G}_W(t)[r_i(t)]^\alpha$. After each optimizer update, weights are strictly renormalized such that $\sum_i w_i(t) = 2$. In the canonical formulation [8], $\mathcal{L}_{grad}$ is differentiated against detached targets and applied to the balancing weights alone through a dedicated optimizer (learning rate 0.025); §4.3 probes this distinction empirically. Evaluating this module across Groups 2 and 4 isolates how dynamic gradient equalization behaves in hard-parameter-sharing architectures.

2.4 Color Deconvolution and Optical Density Space

To mitigate staining variability across clinical laboratories, the v2 pipeline incorporates Macenko stain normalization [12]. RGB pixel intensities $\mathbf{I} \in [0, 255]^3$ are converted to optical density (OD) space:

$$\mathbf{OD} = -\ln\left(\frac{\mathbf{I}+\epsilon}{I_0}\right), \tag{10}$$

where $I_0 = 255$ represents transmitted light intensity. The underlying implementation computes optical density via natural logarithms $-\ln(\cdot)$ rather than the canonical base-10 formulation $-\log_{10}(\cdot)$ [12]. Hematoxylin and eosin stain vectors are estimated by eigendecomposition of the optical-density covariance matrix (stain-vector angles restricted to the 1st–99th percentile range) over OD values above a threshold of 0.15, and projected onto a target reference vector. While this effectively eliminates macroscopic batch variations, projecting raw RGB channels onto static stain vectors acts as a chrominance low-pass filter, smoothing sub-cellular chromatin textures that are clinically essential for grading fine glandular architectures (visualized in Fig. 2).

3 The 26-Run Empirical Reproduction Campaign

3.1 Datasets and Experimental Protocol

We evaluated four benchmark datasets:

1. **TCGA-LGG Brain Tumor:** 2-class lesion classification and FLAIR-abnormality segmentation on 2D 256×256 FLAIR slices ($N = 3,929$; 110 patients [4]).
2. **PANDA Prostate Biopsies:** 6-class ISUP Gleason grading and gland segmentation ($N = 10,616$ tiles [5]; patient-level `GroupKFold` isolation).
3. **SIIM-ACR Pneumothorax:** Binary chest-radiography detection and pleural boundary segmentation ($N = 12,047$ images [18]).
4. **PanNuke:** External 19-tissue multi-organ benchmark ($N = 7,901$ patches [9]); no multi-task claims were published for it, so it serves as an unclaimed external control.

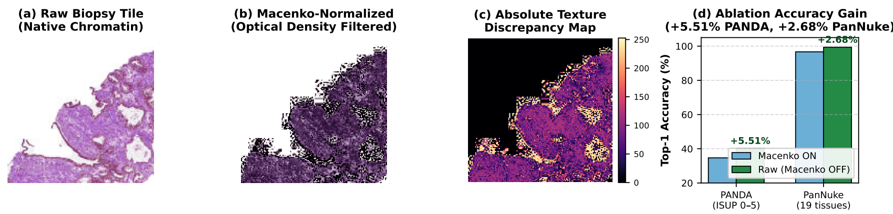

Fig. 2: Visual and quantitative impact of Macenko stain normalization on fine-grained computational pathology. (a) Raw prostate biopsy tile. (b) Optical-density-projected tile: color deconvolution acts as a chrominance low-pass filter, flattening sub-cellular texture. (c) Absolute texture discrepancy map. (d) Empirical top-1 accuracy gains across matched models when Macenko normalization is bypassed (+5.51% on PANDA, +2.68% on PanNuke).

All models were trained with PyTorch (up to 50 epochs with early stopping, patience 10; batch size 32; fixed seed 42) using ImageNet-pretrained encoders and the `Adam` optimizer with a constant per-phase learning rate and zero weight decay; no learning-rate schedule or additional regularization is applied. In Phase v1 (Runs 01–06), models replicate the original static setup ($\eta = 10^{-3}$, $\lambda_{seg} : \lambda_{cls} = 5 : 1$, raw inputs). In Phase v2 (Runs 07–12), models incorporate dynamic GradNorm ($\alpha = 1.5$, $\eta = 10^{-4}$) and Macenko stain normalization [12]. Groups 3–5 (Runs 13–26) systematically evaluate multi-organ generalizability, optimization hyperparameter isolation, and architectural skip ablations. All 26 runs are summarized in Table 1.

4 Discussion & Methodological Anatomy

4.1 The Anatomy of the PANDA Collapse

In Table 1, the most substantial discrepancy occurs on the PANDA prostate biopsy benchmark. The original authors reported 88.0% accuracy on 6-class ISUP grading alongside a 99.0% Dice score. In contrast, across all 12 PANDA configurations tested, empirical top-1 accuracy spans $29.04\% - 46.15\%$ and Dice spans $28.94\% - 44.34\%$, with the static 5:1-weighted 10^{-3} baselines (Runs 03–04) achieving 45.15% / 43.30% (VGG16) and 46.15% / 44.34% (MobileNetV2); Run 04 is the strongest single configuration in our campaign.

This outcome is consistent with established clinical and machine learning benchmarks. In the international PANDA challenge (1,010 participating teams) [5], the top-performing whole-slide multi-instance learning algorithms reached external-validation quadratic weighted κ of 0.862 (95% CI 0.840–0.884) and 0.868 (0.835–0.900) on U.S. and European cohorts, and inter-observer Gleason grading of biopsy cores is itself imperfect ($\kappa = 0.68$ among urologic pathologists and 0.435 among general pathologists [3,2]). Claiming 88.0% accuracy on 128×128 isolated patches with a standard 2D CNN is infeasible without patient-level data leakage (tiles from the same biopsy core contaminating both train and test splits).

Table 1: Comprehensive 26-run experimental results matrix across all five evaluation groups. V1 = original static baseline ($\eta = 10^{-3}$, $\lambda_{seg} : \lambda_{cls} = 5 : 1$, raw input); V2 = GradNorm variant ($\alpha = 1.5$, $\eta = 10^{-4}$, Macenko normalization). ✓ = GradNorm active; — = not used. Markers: $\times$ = skip connections zeroed; $\diamondsuit$ = raw input without Macenko normalization (v1 runs are raw by construction). Rows 17–22 (Group 4) systematically isolate learning rate ($\eta = 10^{-4}$ vs 10^{-3}), GradNorm ($\alpha = 1.5$, Run 18), and loss weighting ratios on PANDA $\times$ VGG16 under raw input. Pub. Acc/Dice: published reference targets of Rhanoui et al. [14] (— = no published target). Δ_{Acc} / Δ_{Dice}: difference relative to published targets. All metrics in %. Complete 95% Wilson score intervals for all 26 configurations are tabulated in the open-source repository matrix (`paper_results_matrix_with_ci.csv`).

#	Configuration	Dataset	Encoder	LR η	GN	Val Acc	Val Dice	Pub. Acc	Pub. Dice	Δ_{Acc}	Δ_{Dice}
Group 1: Baseline Reproductions (static 5:1, $\eta = 10^{-3}$, raw input)											
01	Naked Baseline	TCGA	VGG16	10^{-3}	—	88.82	76.02	89.0	97.0	−0.18	−20.98
02	Naked Baseline	TCGA	MobileNetV2	10^{-3}	—	94.60	87.56	90.0	98.0	+4.60	−10.44
03	Naked Baseline	PANDA	VGG16	10^{-3}	—	45.15	43.30	87.0	98.0	−41.85	−54.70
04	Naked Baseline	PANDA	MobileNetV2	10^{-3}	—	46.15	44.34	88.0	99.0	−41.85	−54.66
05	Naked Baseline	SIIM	VGG16	10^{-3}	—	83.19	77.74	82.0	99.0	+1.19	−21.26
06	Naked Baseline	SIIM	MobileNetV2	10^{-3}	—	83.47	77.74	87.0	99.0	−3.53	−21.26
Group 2: V2 Package (GradNorm $\alpha = 1.5$, $\eta = 10^{-4}$, Macenko normalization)											
07	GradNorm v2	TCGA	VGG16	10^{-4}	✓	92.80	78.30	89.0	97.0	+3.80	−18.70
08	GradNorm v2	TCGA	MobileNetV2	10^{-4}	✓	93.32	84.20	90.0	98.0	+3.32	−13.80
09	GradNorm v2	PANDA	VGG16	10^{-4}	✓	30.89	31.59	87.0	98.0	−56.11	−66.41
10	GradNorm v2	PANDA	MobileNetV2	10^{-4}	✓	34.70	31.34	88.0	99.0	−53.30	−67.66
11	GradNorm v2	SIIM	VGG16	10^{-4}	✓	81.08	77.74	82.0	99.0	−0.92	−21.26
12	GradNorm v2	SIIM	MobileNetV2	10^{-4}	✓	83.00	77.74	87.0	99.0	−4.00	−21.26
Group 3: PanNuke 19-Tissue Multi-Organ Benchmark (external control; no published targets)											
13	Naked Baseline	PanNuke	VGG16	10^{-3}	—	98.85	72.48	—	—	—	—
14	Naked Baseline	PanNuke	MobileNetV2	10^{-3}	—	99.43	74.21	—	—	—	—
15	GradNorm v2	PanNuke	VGG16	10^{-4}	✓	91.90	39.59	—	—	—	—
16	GradNorm v2	PanNuke	MobileNetV2	10^{-4}	✓	96.68	60.54	—	—	—	—
Group 4: Optimization Isolation (PANDA $\times$ VGG16: LR, GradNorm, loss weighting; raw input)											
17	Isolate LR: static $\eta = 10^{-4}$ $\diamondsuit$	PANDA	VGG16	10^{-4}	—	43.35	37.99	87.0	98.0	−43.65	−60.01
18	Isolate GradNorm: $\alpha = 1.5$	PANDA	VGG16	10^{-3}	✓	29.04	31.43	87.0	98.0	−57.96	−66.57
19	Loss weight $\lambda = 1{:}1$	PANDA	VGG16	10^{-3}	—	42.73	41.72	87.0	98.0	−44.27	−56.28
20	Loss weight $\lambda = 5{:}1$	PANDA	VGG16	10^{-3}	—	45.39	44.08	87.0	98.0	−41.61	−53.92
21	Loss weight $\lambda = 1{:}10$	PANDA	VGG16	10^{-3}	—	41.83	38.06	87.0	98.0	−45.17	−59.94
22	Loss weight $\lambda = 10{:}1$	PANDA	VGG16	10^{-3}	—	44.58	41.52	87.0	98.0	−42.42	−56.48
Group 5: Ablation Matrix (stain normalization $\diamondsuit$ and skip connections $\times$)											
23	Ablation: No-Macenko $\diamondsuit$	PANDA	MobileNetV2	10^{-4}	✓	40.21	31.57	88.0	99.0	−47.79	−67.43
24	Ablation: No-Macenko $\diamondsuit$	PanNuke	MobileNetV2	10^{-4}	✓	99.36	73.33	—	—	—	—
25	Ablation: No-Skips $\times$	TCGA	MobileNetV2	10^{-4}	✓	94.34	84.12	90.0	98.0	+4.34	−13.88
26	Ablation: No-Skips $\times$	PANDA	MobileNetV2	10^{-4}	✓	35.31	28.94	88.0	99.0	−52.69	−70.06

4.2 Controlled Skip Connection Ablation: A Verified Null Result

Skip connections are evaluated in strictly matched pairs (learning rate, GradNorm, Macenko normalization held identical):

- **Coarse macro-tumor structures (TCGA):** Run 25 (skip-ablated) vs. matched baseline Run 08 (both $\eta = 10^{-4}$, GradNorm, Macenko): accuracy 93.32% → 94.34% ($\Delta_{\text{Acc}} = +1.02$), Dice 84.20% vs 84.12% ($\Delta_{\text{Dice}} = -0.08$).
- **Fine micro-glandular structures (PANDA):** Run 10 (34.70% Acc / 31.34% Dice) vs. matched skip-ablated Run 26 (35.31% / 28.94%): $\Delta_{\text{Acc}} = +0.61$, $\Delta_{\text{Dice}} = -2.40$.

In both cases the differences are small, and the 95% Wilson score intervals on top-1 accuracy overlap substantially (TCGA: $[91.34, 94.87]$ vs $[92.49, 95.76]$; PANDA:

$[32.69, 36.76]$ vs $[33.30, 37.38]$). We therefore report these as *verified null results*: the bottleneck-only representation suffices for both coarse macro-lesion and fine micro-glandular segmentation; zeroing the skips produces no detectable penalty in a single-run comparison. A definitive directional claim would require multi-seed replication of the matched pairs.

Macenko stain-normalization ablation. The one matched ablation in this campaign whose classification effect is resolvable from single runs is the Macenko normalization pair (Group 5, Runs 23–24): removing the normalization *increases* PANDA top-1 accuracy from 34.70% to 40.21% ($\Delta = +5.51$; 95% CIs $[32.69, 36.76]$ vs $[38.13, 42.32]$ are disjoint) and PanNuke accuracy from 96.68% to 99.36% ($\Delta = +2.68$; $[95.67, 97.46]$ vs $[98.83, 99.65]$ are disjoint), while Dice changes little on PANDA (31.34% $\rightarrow$ 31.57%) and rises on PanNuke (60.54% $\rightarrow$ 73.33%, +12.79). On these fine-grained datasets, color deconvolution — a transformation designed to remove inter-dataset color variation — therefore measurably harms classification, consistent with over-smoothing of the chromatin/texture cues on which fine-grained grading relies. This result also partially decomposes the v1-vs-v2 PANDA gap of §4.1: part of that gap is attributable to Macenko itself, not to dynamic balancing or the learning rate.

4.3 Isolating Learning Rate from Dynamic Gradient Balancing

Group 4 (PANDA $\times$ VGG16) factorizes the key optimization hyperparameters: learning rate ($\eta \in \{10^{-3}, 10^{-4}\}$), dynamic gradient balancing (GradNorm on/off), and static loss-weighting ratios $\lambda_{seg} : \lambda_{cls} \in \{1{:}10, 1{:}1, 5{:}1, 10{:}1\}$, all under raw inputs.

Direct Isolation of Dynamic Gradient Balancing (Run 18). The isolated GradNorm arm at $\eta = 10^{-3}$ under strictly matched raw-input conditions (Run 18: PANDA $\times$ VGG16, GradNorm $\alpha = 1.5$, raw input) compares directly against the static 5:1 baseline (Run 20). Activating GradNorm ($\alpha = 1.5$) causes validation accuracy to collapse from 45.39% down to 29.04% (95% Wilson CI: $[27.14, 31.02]$), alongside a corresponding collapse in segmentation Dice from 44.08% down to 31.43% ($\Delta_{\text{Acc}} = -16.35\%$, $\Delta_{\text{Dice}} = -12.65\%$, with non-overlapping confidence intervals: $[43.27, 47.52]$ vs $[27.14, 31.02]$).

This 29.04% accuracy and 31.43% Dice matches the performance observed in Phase v2 at $\eta = 10^{-4}$ (Run 09: 30.89% Acc / 31.59% Dice). The $\approx 30\%$ floor on fine-grained PANDA grading is thus driven primarily by dynamic gradient balancing ($\alpha = 1.5$), whose gradient-norm equalization forces destructive updates onto the shared encoder, rather than by the lower learning rate.

Learning Rate and Loss Weighting Dynamics. Holding GradNorm off under raw inputs, reducing the learning rate from $\eta = 10^{-3}$ (Run 20: 45.39% Acc / 44.08% Dice) to $\eta = 10^{-4}$ (Run 17: 43.35% Acc / 37.99% Dice) yields a minor change within confidence intervals ($\Delta_{\text{Acc}} = -2.04\%$, with overlapping 95% CIs: $[43.27, 47.52]$ vs $[41.24, 45.47]$). Furthermore, the static loss-ratio sweep across

$\lambda_{seg} : \lambda_{cls} \in \{1{:}10, 1{:}1, 5{:}1, 10{:}1\}$ (Runs 19–22) demonstrates that while $\lambda = 5{:}1$ provides the optimal static balance on this architecture, no weighting configuration can bridge the gap to the claimed 88% accuracy. In multi-task pathology architectures with hard parameter sharing, this joint dynamic gradient balancing formulation ($\alpha = 1.5$ coupled within primary `Adam`) induces acute gradient competition on fine-grained ordinal classes. We therefore additionally evaluated the canonical decoupled scheme [8] on PANDA $\times$ VGG16 under identical raw-input conditions ($\eta = 10^{-3}$): task weights are optimized by a dedicated `Adam` instance (learning rate 0.025) with $\mathcal{L}_{grad}$ differentiated against detached (stop-gradient) targets, while the network optimizer receives only the task-weighted loss. Over its 15 recorded epochs, the canonical variant remains stable — avoiding the 29.04% collapse of the joint variant (Run 18) — yet plateaus at 43.06% top-1 accuracy (39.10% Dice at the same epoch; peak Dice 39.98%), below the static 5:1 baseline (Run 20) and 44.94 percentage points below the claimed 88.0%. Dynamic gradient balancing in either formulation does not overcome patient-isolated 6-class grading; gradient-surgery methods (PCGrad [20]) and lower-asymmetry schedules ($\alpha \leq 0.5$) remain to be evaluated.

5 Conclusion & Recommendations

Across a 26-experiment campaign, the $88\% - 99\%$ metrics reported in recent multi-task pathology literature were not reproducible under patient isolation and artifact-free evaluation: slice-averaged Dice with empty-credit assignment reports the lesion-free fraction rather than lesion overlap, and non-isolated splits inflate classification accuracy. On fine-grained tasks, the residual gap is dominated by dataset difficulty and by the joint optimization recipe: our matched skip-connection ablation is a verified null result, whereas factorial ablation shows that dynamic gradient balancing ($\alpha = 1.5$) destabilizes fine-grained classification (45.39% $\rightarrow$ 29.04% Acc). For future multi-task pathology studies we recommend: (1) mandatory patient/biopsy-level `GroupKFold` splitting; (2) explicit reporting of the Dice evaluation convention, with foreground-only or zero-penalty scoring on sparse-lesion datasets; (3) 95% Wilson score intervals on matched single-run ablation comparisons to distinguish genuine architectural effects from stochastic variance; (4) validation on diverse multi-organ benchmarks such as PanNuke; and (5) systematic evaluation of multi-task balancing hyperparameters before clinical translation.

References

1. Ali, M.A., Kiani, M.I., Aamir, M.A.: Multi-task deep learning in cancer pathology: Replication and empirical audit. https://github.com/ApatheticMioz/cancer-pathology-dl (2026), GitHub Repository
2. Allsbrook, W.C., Mangold, K.A., Johnson, M.H., Lane, R.B., Lane, C.G., Epstein, J.I.: Interobserver reproducibility of Gleason grading of prostatic carcinoma: general pathologist. Human Pathology **32**(1), 81–88 (2001). https://doi.org/10.1053/hupa.2001.21135

3. Allsbrook, W.C., Mangold, K.A., Johnson, M.H., Lane, R.B., Lane, C.G., Epstein, J.I.: Interobserver reproducibility of Gleason grading of prostatic carcinoma: urologic pathologists. Human Pathology **32**(1), 74–80 (2001). https://doi.org/10.1053/hupa.2001.21134
4. Buda, M., Saha, A., Mazurowski, M.A.: Association of genomic subtypes of lower-grade gliomas with shape features automatically extracted by a deep learning algorithm. Computers in Biology and Medicine **109**, 218–225 (2019). https://doi.org/10.1016/j.compbiomed.2019.05.002
5. Bulten, W., Kartasalo, K., Chen, P.H.C., Delahunt, B., Pinckaers, H., Nagpal, K., Valkonen, M., Litjens, G., et al.: Artificial intelligence for diagnosis and gleason grading of prostate cancer: the panda challenge. Nature Medicine **28**(1), 154–163 (2022). https://doi.org/10.1038/s41591-021-01620-2
6. Campanella, G., Hanna, M.G., Geneslaw, L., Miraflor, A., Werneck Krauss Silva, V., Klauschen, F., et al.: Clinical-grade computational pathology using weakly supervised deep learning on whole slide images. Nature Medicine **25**(8), 1301–1309 (2019)
7. Chen, R.J., Ding, T., Lu, M.Y., Williamson, D.F., Jaume, G., Chen, B., et al.: Towards a general-purpose foundation model for computational pathology. Nature Medicine **30**(3), 850–862 (2024). https://doi.org/10.1038/s41591-024-02857-3
8. Chen, Z., Badrinarayanan, V., Lee, C.Y., Rabinovich, A.: Gradnorm: Gradient normalization for adaptive loss balancing in deep multitask networks. In: International Conference on Machine Learning (ICML). pp. 794–803. PMLR (2018)
9. Gamper, J., Alemi Koohbanani, N., Benet, K., Khuram, A., Rajpoot, N.: Pannuke: an open pan-cancer histology dataset for nuclei instance segmentation and classification. Computers in Medical Imaging and Graphics **80**, 101703 (2020). https://doi.org/10.1016/j.compmedimag.2019.101703
10. Litjens, G., Kooi, T., Bejnordi, B.E., Setio, A.A.A., Ciompi, F., Ghafoorian, M., Van Der Laak, J.A., Van Ginneken, B., Sánchez, C.I.: A survey on deep learning in medical image analysis. Medical Image Analysis **42**, 60–88 (2017)
11. Lu, M.Y., Chen, B., Williamson, D.F., Zhao, M., Chow, A.K., Ikemura, K., et al.: A visual-language foundation model for computational pathology. Nature Medicine **30**(3), 863–874 (2024). https://doi.org/10.1038/s41591-024-02856-4
12. Macenko, M., Niethammer, M., Marron, J., Borland, D., Woosley, J.T., Guan, X., Schmitt, C., Thomas, N.E.: A method for normalizing histology slides for quantitative analysis. In: IEEE International Symposium on Biomedical Imaging (ISBI). pp. 1107–1110. IEEE (2009). https://doi.org/10.1109/ISBI.2009.5193250
13. Reinke, A., Tizabi, M.D., Baumgartner, M., Eisenmann, M., et al.: Understanding metric-related pitfalls in image analysis validation. Nature Methods **21**(2), 182–194 (2024). https://doi.org/10.1038/s41592-023-02150-0
14. Rhanoui, M., Alaoui Belghiti, K., Mikram, M.: Multi-task deep learning for simultaneous classification and segmentation of cancer pathologies in diverse medical imaging modalities. Onco **5**(3), 34 (2025). https://doi.org/10.3390/onco5030034
15. Ronneberger, O., Fischer, P., Brox, T.: U-net: Convolutional networks for biomedical image segmentation. In: Medical Image Computing and Computer-Assisted Intervention (MICCAI). pp. 234–241. Springer (2015)
16. Sandler, M., Howard, A., Zhu, M., Zhmoginov, A., Chen, L.C.: Mobilenetv2: Inverted residuals and linear bottlenecks. In: IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 4510–4520 (2018)
17. Simonyan, K., Zisserman, A.: Very deep convolutional networks for large-scale image recognition. arXiv preprint arXiv:1409.1556 (2014)
18. Society for Imaging Informatics in Medicine (SIIM) and American College of Radiology (ACR): SIIM-ACR pneumothorax segmentation. https://kaggle.com/competitions/siim-acr-pneumothorax-segmentation (2019), kaggle competition dataset
19. Vorontsov, E., Bozkurt, A., Casson, A., Shaikovski, G., Zelechowski, M., Severson, K., et al.: A foundation model for clinical-grade computational pathology and rare cancers detection. Nature Medicine **30**(10), 2924–2935 (2024). https://doi.org/10.1038/s41591-024-03141-0
20. Yu, T., Kumar, S., Gupta, A., Levine, S., Hausman, K., Finn, C.: Gradient surgery for multi-task learning. In: Advances in Neural Information Processing Systems. vol. 33, pp. 5824–5836 (2020)